DS605: Fundamentals of Machine Learning

Lab Assignment - 7

Image Representation, PCA, Eigenfaces and Face Recognition

Dataset: AT&T Database of Faces (ORL)
Submission: Public GitHub repository link only.

Objective

Represent images numerically, reduce their dimensionality using PCA/eigenfaces, perform face matching in the reduced space, and compare the result with traditional machine-learning classifiers.

Part A - Image Representation

- Read the face images as grayscale pixel matrices, inspect their dimensions and intensity values, flatten each image into a feature vector, and create the complete image matrix with subject labels.
- Create a reproducible train-test split with images from every subject represented in both sets.

Part B - PCA, Eigenfaces and Compression

- Compute the mean face and center the training data. Use covariance/eigendecomposition and SVD to obtain the principal directions and compare the two approaches.
- Plot cumulative explained variance and retain the minimum number of components needed to preserve at least 90% variance. Reshape selected principal components back into images to visualize eigenfaces.
- Project faces into PCA space, store the reduced representation, reconstruct sample faces, and compare reconstruction quality and storage with the original pixel representation.

Part C - Recognition in PCA Space

- For each test image, generate its PCA representation and identify the closest stored training face using a suitable distance measure such as Euclidean distance.
- Report recognition accuracy, inspect a few correct and incorrect matches, and compare recognition using raw pixels with recognition using the reduced PCA representation.

Part D - Traditional Machine Learning

- Use the PCA features to train at least two traditional classifiers such as KNN, Logistic Regression, or SVM, and compare them with the similarity-based baseline.
- Experiment with the number of retained components and suitable model settings to improve recognition while keeping the representation compact.

Expected Workflow

Images -> Grayscale Pixel Matrix -> Flatten -> Mean Centering -> PCA / SVD -> Explained Variance -> Eigenfaces -> Reduced Representation -> Reconstruction -> Similarity Matching -> ML Classification

Important Requirement

- PCA/eigenface calculations must be implemented using NumPy linear-algebra operations rather than Scikit-learn PCA. Image-reading libraries may be used for file input; Scikit-learn may be used only for the final traditional-ML comparison.

Deliverables (GitHub Only)

- Public repository containing runnable code/notebook, explained-variance and eigenface figures, reconstruction examples, recognition results, ML comparison table, and a README describing the approach and observations.
- Submit only the public GitHub repository link.